



UI / UX DESIGN CASE STUDY

Modernizing Contractor Registration & Renewal

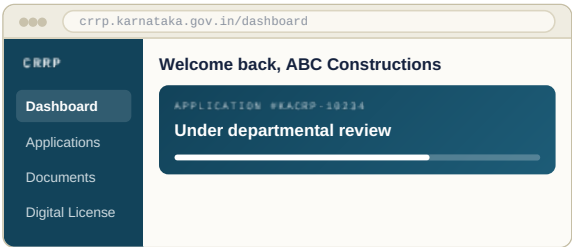
Designing a unified, secure and mobile-responsive digital experience that turns a paper-dependent government workflow into one connected service — for civil contractors across Karnataka.

ROLE
UI / UX Designer

PLATFORM
Responsive Web App

DOMAIN
GovTech · Public Services

CLIENT
Karnataka PWD



"Make contractor registration feel like one service, not a collection of disconnected systems."



The Challenge

A **digital portal built around a paper-based process**. The existing Civil Contractor Registration & Renewal Portal was developed in 2014–15. Contractors could technically apply online, but the workflow still leaned on printed applications, physical files, bank correspondence and manual verification at every stage.

Separate portals existed for different zones and contractor classes — so the “digital” system still behaved like a fragmented, office-by-office process. The opportunity was clear:

Transform a fragmented, paper-dependent registration process into one connected digital experience.

The Problem

For contractors

- Multiple portals to keep track of
- Complicated registration & renewal steps
- Registration numbers that are hard to recall
- Manual processes for certain contractor classes
- Little to no visibility into application status
- Physical documentation still required

For department teams

- Paper files running alongside online applications
- Difficult, scattered document management
- Inconsistent workflows across offices
- Manual task assignment
- Limited user & access management
- Challenging license-status handling

The Goal

01 One place — bring registration, renewal and upgrades together.

02 One identity — give every contractor a single account across their lifecycle.

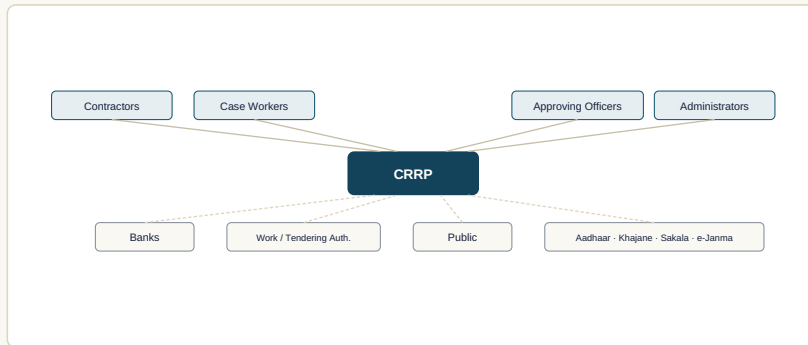
03 Guided completion — break complex forms into understandable steps.

04 Transparent progress — make application status and next actions visible.

05 Digital by default — reduce unnecessary paper, manual checks and physical credentials.

Understanding the Ecosystem

CRRP isn't just a contractor-facing portal — it connects everyone involved in registration and verification. Designing the interface meant designing the system around it first.



Information Architecture — Contractor Portal

- Dashboard
- My Applications
- New Registration
- Renewal / Upgrade License
- Documents
- Payments
- Appointments
- Digital License
- Notifications · Help

Information Architecture — Department Portal

- Dashboard
- Applications
- Tasks
- Contractors
- Documents
- Workflow
- Users & License Mgmt.
- Reports
- Settings

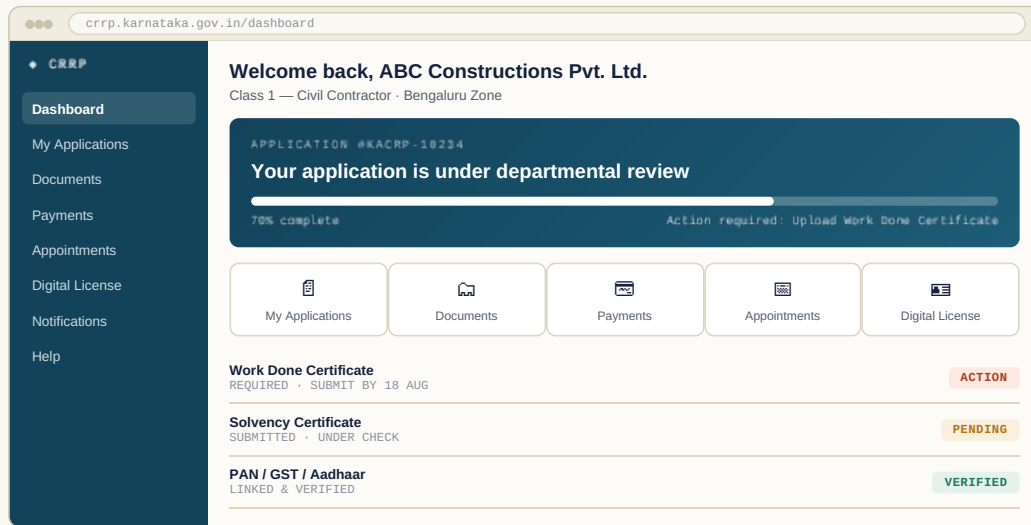
The Registration Journey

Ten stages, simplified into one guided path — from account creation to an approved, verifiable license.



One Question, Answered First

The dashboard is built around a single question a contractor actually has: "what do I need to do next?" Instead of a menu of modules, the first thing they see is their live application status, the one action required of them, and a clear path to everything else.



Design rationale. Status, not navigation, leads. The hero card always states where the application stands and what the contractor must personally do — never a generic "in progress." Quick-access tiles keep the rest of the portal one tap away without competing with the primary task, so the dashboard scales from a first-time registration to years of renewals.

Making Complex Forms Easier

Government registration asks for a large amount of information at once. Rather than one long form, it's broken into focused sections — Business Details → Applicant Details → Technical Person → Work Experience → Financial Information → Documents → Review & Submit — each explaining what's needed, why, and what happens next.

crp.karnataka.gov.in/apply/technical-person

1

Business

2

Applicant

3

Technical person

4

Work experience

5

Financial

6

Documents

7

Review

Technical person name

Ramesh K. Gowda

A named technical person is required for Class 1–2 contractor categories. This links responsibility for site work to a verifiable individual, and speeds up department review.

Qualification

Diploma in Civil Engineering

Experience (years)

9

License / ID number

TP-KA-04521

Upload identity proof

Drag file or browse — PDF, JPG up to 5 MB

— DOCUMENT MANAGEMENT

From a List of File Names to a Clear Checklist

Every document tells the contractor what it is, why it's required, and what happens next — replacing an undifferentiated upload list with a real checklist.

DOCUMENT	WHY IT'S NEEDED	STATUS
PAN	Business identity verification	VERIFIED
GST	Tax registration check	VERIFIED
Aadhaar	Applicant identity link	VERIFIED
Solvency Certificate	Financial standing for class	REQUIRED
Work Done Certificate	Prior project evidence	REQUIRED
Technical Person Document	Qualification proof	SUBMITTED

Application Tracking

Contractors previously had no way to see where their application stood. The tracker gives every application a visible timeline and a single, current-action line — so "no news" never means "no idea."

●●●

crrp.karnataka.gov.in/applications/KACRP-10234

APPLICATION #KACRP-10234

● Application submitted

03 AUG 2026

● Documents verified

06 AUG 2026

● Payment confirmed

06 AUG 2026

● Department review

IN PROGRESS

● Final approval

NOT STARTED

● License issued

NOT STARTED

Current action: none required. Your application is being reviewed by the registering authority.

●●●

crrp.karnataka.gov.in/appointments/book

BOOK VERIFICATION APPOINTMENT

Verification office

KPWD — Bengaluru

Select a date

MON TUE WED THU FRI

Select a time

10:00 AM 11:30 AM 2:00 PM 3:30 PM

Appointment booked successfully. Bring the original documents listed in your application.

Original document verification remains part of the process, so the digital experience makes the physical step predictable too — a confirmed office, date and time, with a clear reminder of what to bring.

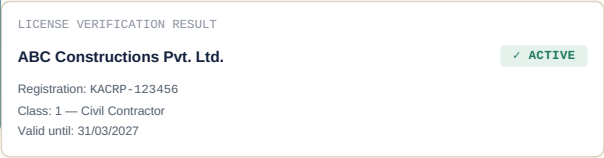
From Smart Card to QR License

The proposal replaces the physical MIFARE smart-card approach with a QR-based license — simpler to issue, simpler to verify, and impossible to leave at home.



Instant Verification

Scanning the license opens a read-only, tamper-proof confirmation — giving tendering authorities and the public a faster way to trust contractor credentials without relying on a physical card.



A Simpler Contractor Identity

Existing registration numbers combined zone, class, category, serial number and year into one string. The proposal replaces this with a single alphanumeric identifier, closer to a GST or PAN format.

CBS/C1/CIVIL/1234/2015 → KACRP-123456

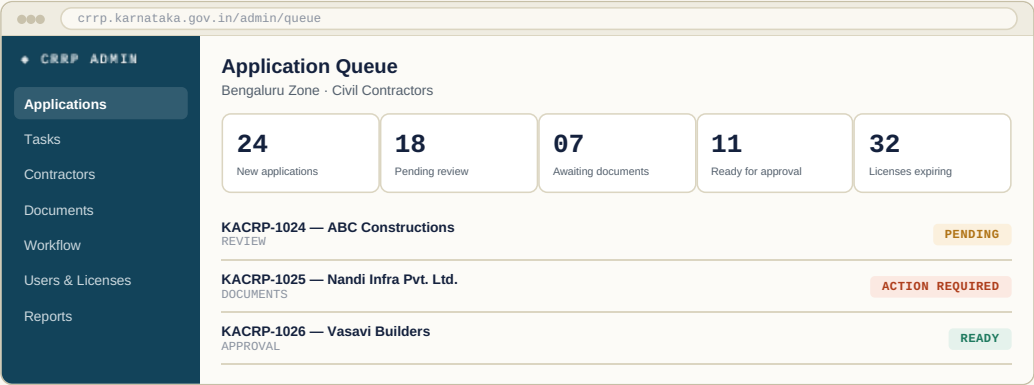
Secure by Design

Multi-factor authentication adds a layer of protection to every portal login, without adding unnecessary friction to the experience.



Designing for the Department

The contractor experience is only one side of the product. For departmental users, the interface shifts from forms-first to **work-first** — an operational queue built around what needs attention today.



Design rationale. A case worker's day starts with "what's in my queue," not Status pills reuse the same visual language as the contractor portal — "search for a contractor." The workspace leads with counts and priority, not a blank search bar. **VERIFIED**, **PENDING**, **ACTION** — so state means the same thing on both sides of the service.

Responsive by Default

The modernized portal is designed to adapt to context, not just screen size — each device supports a different moment in the contractor's day.

Desktop

More information density and full workflow visibility — for form-filling and document review.

Tablet

Optimized forms and document management — for on-site or office-counter use.

Mobile

Prioritizes status, tasks, notifications, documents, appointments and the digital license.

Before → After

Multiple portals	→	One unified portal
Paper + online	→	End-to-end digital
Complex registration number	→	Simplified unique ID
Password-only login	→	Multi-factor authentication
Physical smart card	→	QR digital license
Scattered documents	→	Central document management
Limited tracking	→	Application timeline
Difficult license verification	→	Instant QR verification

The Outcome

Contractors

A simpler way to register, renew, track and manage licenses.

Government teams

A centralized workspace for applications, documents, tasks and workflows.

Verification authorities

Faster access to trustworthy contractor information.

The ecosystem

A connected platform able to integrate with other government services.

WHAT I LEARNED

Designing government services is less about making forms beautiful, and more about making complexity understandable.

The biggest UX opportunity here was never a visual redesign — it was understanding the entire service ecosystem and connecting its pieces:



By designing around that journey, CRRP can move from a collection of legacy workflows toward a connected, transparent and scalable public service.

From paperwork to progress.

CRRP modernization reimagines contractor registration as a connected digital service — bringing registration, verification, payments, workflows and licensing into one experience.

SOURCE NOTE — This case study is based on the CRRP modernization proposal, which describes current-state problems, requirements and implementation scope. It does not include post-launch UX metrics or usability-test findings. Interface screens in this document are original design mockups created to illustrate the proposed experience, not final production UI.